# EXP 1 - ARDUINO UNO-BASED SMART HOME SAFETY AND OCCUPANCY MONITORING SYSTEM USING GAS, PIR AND ULTRASONIC SENSORS

**CODE:**

```
#define MQ2_PIN A0
#define TRIG_PIN 9
#define ECHO_PIN 8
#define PIR_PIN 2
#define LED_PIN 10
#define BUZZER_PIN 11

int gasValue;
int gasThreshold = 400;

long duration;
int distance;

void setup()
{
  pinMode(MQ2_PIN, INPUT);
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(PIR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  Serial.begin(9600);

  Serial.println("SMART DISASTER MONITORING SYSTEM");
  Serial.println("System Initialized");
}
```

```
void loop()
{
  // Read Gas Sensor
  gasValue = analogRead(MQ2_PIN);

  // Read PIR Sensor
  int motion = digitalRead(PIR_PIN);

  // Read Ultrasonic Sensor
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  duration = pulseIn(ECHO_PIN, HIGH, 30000);
  distance = duration * 0.034 / 2;

  Serial.println("-------------------------");

  Serial.print("Gas Level: ");
  Serial.println(gasValue);

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  Serial.print("Motion: ");
  Serial.println(motion ? "Detected" : "Not Detected");
```

```
 // Check Emergency Conditions
 if (gasValue > gasThreshold || motion == HIGH ||
     (distance > 0 && distance < 50))
 {
   digitalWrite(LED_PIN, HIGH);
   digitalWrite(BUZZER_PIN, HIGH);

   Serial.println("WARNING: EMERGENCY DETECTED!");
 }
 else
 {
   digitalWrite(LED_PIN, LOW);
   digitalWrite(BUZZER_PIN, LOW);

   Serial.println("STATUS: NORMAL");
 }

 delay(1000);
}
```

**OUTPUT**

